

AT2 - Real Estate Agency “Easy Property”
Proposal: Free up team capacity, improve employee morale, reduce costs, and ultimately provide better customer service via a machine learning model.

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Introduction

Easy Property is a Real Estate Agency that provides advertising for clients all over the country who are looking to sell their homes. They do this by posting their clients properties, using images, and listing them on multiple listing websites. This is currently a manually driven process and an agent at Easy Property will need to manually review a client's property photos and categories these photo's i.e., bedroom, living room, kitchen etc. so that they can be uploaded effectively to the necessary listing websites.

Feedback from Easy Property agents indicate that the listing websites are not particularly user friendly and the process of categorizing and uploading can take up to 20min for a property. And with 50,000 new properties being added every month this expands to hours and hours of mundane and repetitive work for staff. In fact, this equates to 200,000 hours a year or the equivalent of 145 staff working all year round just processing photos all day.

There is however a way to give time back to Easy Properties staff so that they can focus on greater value add and customer facing work. Deploying an Image Classification machine learning model will enable Easy Property to automatically classify images.

Conservative estimates indicate that Easy Property can achieve a ROI of ~1 by implementing an Image Classification Machine Learning model.

In this proposal we will walk through the business case for change, what are the proposed requirements and project phases to implement a successful Image Classification model to deliver benefits to clients, staff, and the business.

Lastly, we'll recommend the MLOps level that Easy Property should implement as part of their operating model moving forward.

Business Case for change and solution requirements

Before embarking on implementing an AI solution we need to understand:

1. Is there a clear business case for the change. What is the cost and the customer and/or market impact of implementing the change?
2. What are the solution requirements to achieve the business objectives.

To answer these questions, we will look at the 5 key requirements for a successful Machine Learning product¹.

1. Business Value:

From a cost-effective standpoint let's understand how much effort is wasted in the manual process for Easy Property.

New Properties listed per month	=	50,000
Total properties in a year	=	600,000
Effort required in hours /listing	=	0.33
Total effort hours in a year	=	200,000
Average admin assistant wage per hour ²	=	\$32.89
Annualised effort cost	=	\$6,578,000

So, the resourcing cost to categories and load the photos is over \$6.5 million a year.

Now let's assume a conservative case that only 75% of the effort can be recovered through a ML model. That still creates an opportunity of roughly \$4.93 million.

Let's look to understand the potential costs for a ML solution. If we assume that there are on average 10-15 pictures³ for each property that need to be processed. This equates to 500,000 – 750,000 images per a month and 6-9 million images per year.

Based on the above volumes and assuming, Data acquisition and labelling, Model development, Infrastructure and computing resources, Deployment infrastructure and Ongoing maintenance and updates ChatGPT⁴ indicates that the cost can range from hundreds of thousands to couple million.

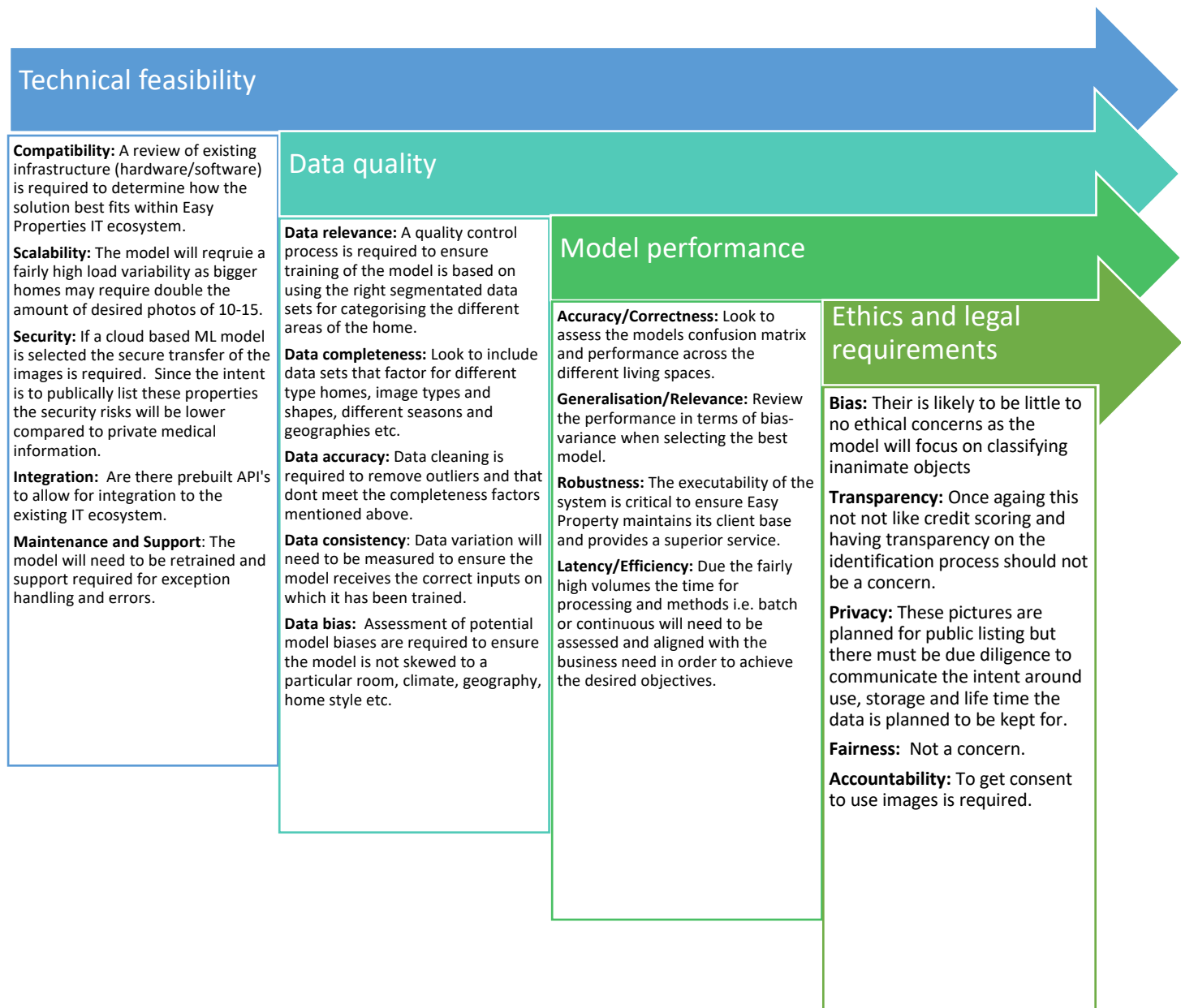
Assuming a conservative costing of \$2.5 million a ROI of 0.97 can still easily be achieved.

This is only looking at the direct labour costs there is still further benefits in terms of:

- **Improved employee satisfaction and engagement as this mundane repetitive work can be removed and they can focus on greater value adding work.**
- **Faster turnaround times for customers with real-time processing not 9-5 and weekdays.**
- **Further competitive advantage for the organisation as the ability to scale digitally will be easier and faster than recruiting and training new staff.**

2. Solution requirements

Beyond the business case the following solution requirements will need to be assessed. A brief overview is provided below. Greater detail and granularity is provided in the CRISP-ML Project Plan to in the following section.



Project Plan (CRISP-ML)

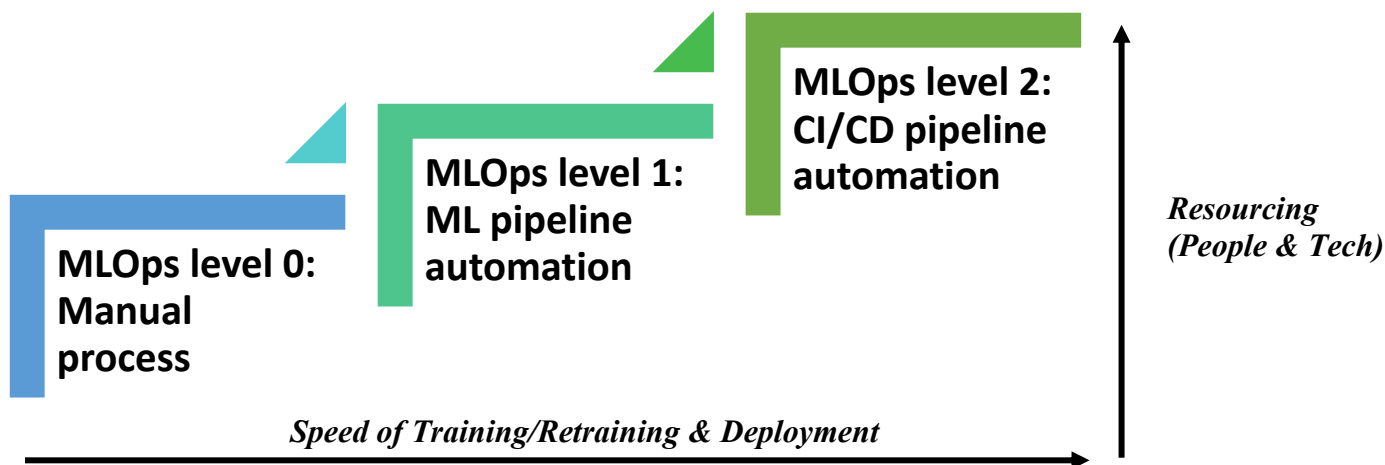
The table on the following pages has been populated based on the ML-Ops, CRISP-ML(Q). The ML Lifecycle Process¹⁰. In the table we will cover the key phases, tasks, and proposed actions to be taken by Easy Property to implement the desired ML solution. We will also provide some high-level indications of Timelines, Effort, Costs and Risks.

CRISP-ML(Q) Phase	Tasks	Proposal	Timeline ⁷	Effort	Cost	Risks
Business and Data Understanding	- Define business objectives	There are 3 business objectives: - Reduce manual effort to free up staff capacity and reduce unit costs. - Lift employment satisfaction by allowing them to work on more value adding tasks. - Reduce customer turnaround time and increase scalability for future growth.	1-2 months	Medium	Low	- Ill defined and non-specific objectives can have a dramatic impact on the overall outcome of ML projects. Defining and managing teams to deliver on the required objective is key. Gartner⁵ states "85% of AI projects will deliver erroneous outcomes due to bias in data, algorithms or the teams responsible for managing them." - The old saying of "Garbage in, Garbage out" rings true. Spending the time and resources to get the inputs into the model is critical in improving the likely hood of an accurate model. - Resourcing for manual exception handling will still be required and manual contingencies may be required if there is a surge on ML processing impacting its latency and efficiency. - This stage is used to assess the likelihood of success and requirements to remove/minimise and/or mitigate the risks. - The proof of concept should be critically reviewed by sponsors and key stakeholders to assess if the solution will achieve the desired outcomes. External parties should also form part of this review to prevent internal bias and should form the final step before signing off to go into product development phase.
	- Translate business objectives into ML objectives	- Reduce the manual effort required to classify photos by a minimum of 75%. - Half the time taken or end-to-end time to classify and load photos on the listing agency platforms. - Statistically significant uplift in staff engagement scores as a result of the project change.				
	- Collect and verify data	Catalogue Data: The required ML data set will need to be explored to define data attributes and understand the current data set types in terms of format, size, colour, location etc. Data profiling: Statistical methods such as average, variance, maximum, minimum and etc on data asset attributes. Visualisation methods will also be completed so that a Data Scientist can interpret the statistical results to identify outliers and rule out spurious correlations that may later impact feature selection. Data verification: Data quality requirements will be established and a manual exception process will be required. Based on the current objectives there is a maximum allowance of 25% on the exception rate. In model testing and verification stage this can be further reduced based on model accuracy and robustness.				
	- Assess the project feasibility	The feasibility is to be assess across multiple dimensions. Research and investigation will be required across 3-12 weeks ⁶ and will need to cover: 1. Technical - What are the hardware, software, licensing and internal/external skill capability requirements for the proposed solution. What are the scalability requirements. 2. Financial - Benefit vs. Cost analysis to determine if their is sufficient ROI. 3. Strategic - Is this project aligned with the strategic vision of the organisation. How will this provide competitive advantage and superior customer and/or staff satisfaction. 4. Risk & Legal Compliance - What are the risk implications and are there any regulatory impacts as a result in implementing the solution.				
	- Create POC	Generate a Proof of Concept perhaps with a strategic partner (trusted listing agency) for a selected category of the data set i.e. like for kitchens and/or bathrooms. Look to leverage off existing model types if available with a focus on proving if an end-to-end solution can be achieved and is viable to move into the true development phase.				
Data Engineering	- Feature selection	Determine what are the most important aspects/features of the images and how do we extract them using the model. This will require a domain expert to complete an analyse using various methods i.e. filter, wrapper and embedded and techniques on the images such as: - Cropping - Grayscale (less data required are we removing important aspects.) - RGB channels - Intensity threshold - Edge detection to determine the futures that deliver the best outcomes at lowest model complexity and cost. Therefore accessing the bias-variance trade-off is critical and will impact on the data selection process.	2-4 months	Medium	Medium	- The failure to use a competent domain expert can result in a substandard model algorithm with reduced accuracy and efficiency. There expertise is required to support the model on interpreting the real world features of the data and understanding the scale and impact of the features. - Poor data preparation will lead to "dirty" models that will impact model effectiveness.
	- Data selection	When selecting the data we will need to do a exploratory analysis. In this we will need to assess: - Distribution of the images categories i.e. number of images for bedrooms, toilets, living spaces etc. - The type and quality of the images. Poor quality images may be excluded from training the model. - The size of the image as this may require standardisation. - The image channel (i.e. 3 for RGB images)				
	- Cleaning data	For our case this will involve removing outliers such as images impacted by high amounts of lighting our outdoor rooms like showers or other living spaces. Another one could be unusual angles like an aerial photo taken of a room from the floor above etc.				
	- Data augmentation	For the image augmentation this will support with expanding the data set and can include adjusting image. - Rotation - Height - Width - Horizontal flip				
	- Data standardization	Normalisation of the data to prevent features from skewing the results based on their scale. This will be less of an issue for image classification				

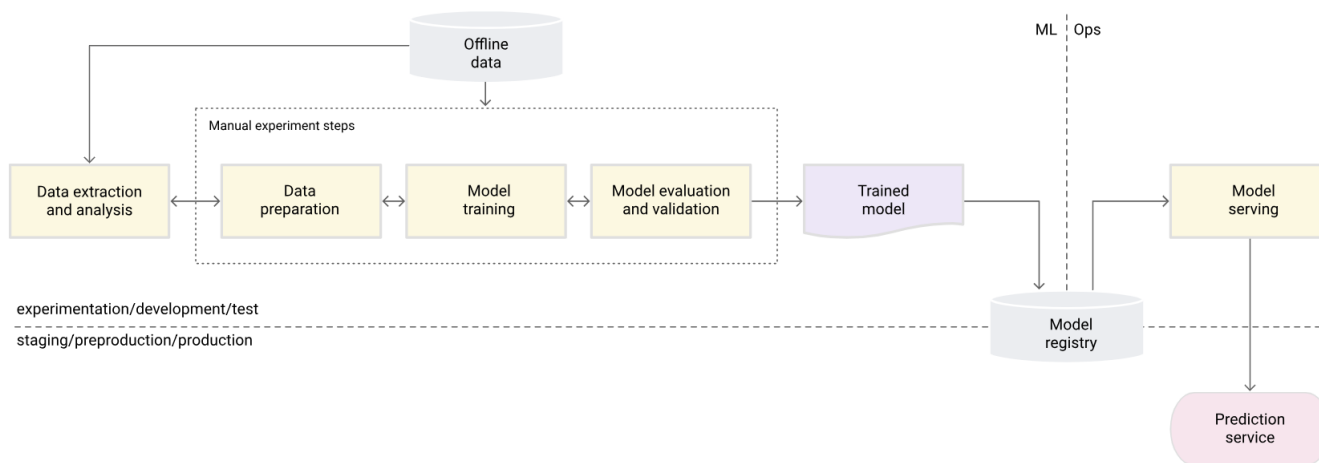
ML Model Engineering	- Define quality measure of the model	The following measures, Accuracy, Precision, Recall, F1 Score and Confusion Matrix will be assessed during the algorithm selection process. Look at using k-fold cross validation to better understand the average performance of the model and standard deviation.	2-6 months	High	High	- Only assessing the performance of one model. The analysis should be completed by a domain expert across multiple models using the defined quality measures to determine the best fit. Options can also be explored to assess the model performance across segmented data sets and multiple models could be used. This will however increase costs and hence there needs to be a cost/benefit analysis to justify the use. - Failure to document the ML Model Engineering processing resulting in outcomes not being able to be replicated if there is a need to replicate in another environment or to license the solution. Business unable to explain how the results are achieved causing it to be viewed as a "black box".
	- ML algorithm selection (baseline selection)	Referencing from a case study by Sumedh Bhagat (Sept, 2021) for "Real Estate Image Classification" ⁸ the following classifications models can be evaluated for testing and accuracy. - Simple Sequential Model - VGG16 Model - ResNet50 Model - EfficientNetB5 Model - Post Training Quantization EfficientNetB5 Model - Xception Model - AlexNet Model The models above include simple sequential models for basic tasks, powerful architectures like VGG16, ResNet50, EfficientNetB5, and Xception for advanced computer vision tasks, and techniques like post-training quantization for memory and computational efficiency. So the intent would be to start from simple and progress to more advanced and computational if needed to achieve the objectives.				
	- Model training	Each model will need to be trained on a subset of the data set i.e. 60%. The training should ensure there is sufficient volume of data and segmented categories at the required quality. Suggest using the 10x rules for data relative to features.				
	- Optional: applying transfer learning (using pre-trained models)	Data labelling can be time consuming and costly so pre-trained supervised models can be used to help label the data before using our model.				
	- Ensemble learning	The model selection, training and evaluation of models in terms of their confusion matrixes will enable the models to be ranked. Ensemble methods can be used to then look to use multiple models and make a determination based on the accuracy of multiple models. This may not be required for this case as the consequence or risk of an error is relatively low. If this was for classifying a medical diagnosis i.e. like cancer then this may be a suitable option.				
	- Documenting the ML model and experiments	- Document assumptions, modelling and analysis so that it can be replicated and explained internally and externally. A model card and registry like Mflow can be used.				
ML Model Evaluation	- Validate model's performance	A separate validation set can be used to fine-tune the model during training, allowing adjustments to hyperparameters, feature selection, or model design before final evaluation on the testing set. Hyperparameter tuning by systematically adjusting the model's hyperparameters by grid search or random search, to optimize performance. Examples include the number of convolutional layers and degree of data augmentation.	1-3 months	Medium	Low	- Not assessing The soft metrics like robustness and explainability can lead to significant deployment issues. Where The model runtime is impacted and The business is unaware of how to fix or tune The model due to its low explainability. - The decision is made to go ahead due to previous sunk cost. The decision to move into the deployment phase must make both future economic and practical sense. This needs to be an objective decision by the project sponsor and they should potentially get a peer or third party to validate the findings and decision.
	- Determine robustness	Metrics like performance, robustness, fairness, scalability, interpretability, complexity, and resource demand will need to be consider. Not only must performance be assessed but also its robustness and explainability. Addressing fairness and privacy will have less impact for our application due to the nature of the problem.				
	- Increase model's explainability	Use of model card and model registry Mflow - https://mlflow.org/docs/latest/model-registry.html#concepts				
	- Make a decision whether to deploy the model	Based on the performance relative to the desired objectives and business case a decision can be made to move into deployment.				
	- Document the evaluation phase	For both the success and failure of the project the evaluation phase needs to be documented so that good results can be repeated and for mistakes root cause analysis done and corrective action taken moving forward.				
Model Deployment	- Evaluate model under production condition	Deployment can under the following methods: - Rest API to serve online predictions - An embedded model to an edge or mobile device - Part of an offline batch prediction system	1-2 months	High	High	- The end-to-end deployment and integration of the system is not ready. - There is no clear governance structure to manage the successful deployment and maintenance of the solution. - RACI is not established on who is accountable to deliver on which part of the deployed solution.
	- Assure user acceptance and usability	Test the solution with selected partners to obtain client feedback and validate customer expectations.				
	- Model governance	- Implement a governance framework that ties into The model monitoring and maintenance phase to support sustainability and control.				
	- Deploy according to the selected strategy (A/B testing, multi-armed bandits)	Implement a Canary deployment by drip feeding targeted client data based on agreement and testing with partners. Once production is stable and performance is at desired levels increase the rate of drip feeding.				
Model Monitoring and Maintenance	- Monitor the efficiency and efficacy of the model prediction serving	Need to access model to course correcting due to drifting. There are two types of drifting namely Concept and Data. Concept is where there is a behavioural or relationship change and data drift is where there is a change in the data (i.e. label or feature drift). The drifting can be from varying changes in the data and real world. It can be: - Abrupt changes like with a change in technology and transition for picture formats. - Gradual changes and the styles and layouts of homes change over time. - Incremental - Recurrent as with the changes in the seasons. - Blip The drifting can be caused based on the fact that the model was trained based on data at a specific time. Therefore looking at the distribution of training data to the real world now can also be a useful to correct against drifting. Hints on drift in starting to identify the root cause is to look at data integrity and outlier monitoring first.	Ongoing service	Low	Low	- No maintenance and model performance declines over time. - Implement a higher MLOps Maturity level that incurs additional costs with limited benefits assuming there is no current criteria to expand on more ML projects.
	- Agree on MLOps level of maturity	Recommended to follow Maturity Level - 0 and is covered in the following section.				

Machine Learning Operations (MLOps) selection process

There are 3 fundamental maturity levels for implementing MLOps. These maturity levels range from 0 to 2 with the maturity level increasing to allow for greater speed of training and deployment of ML solutions. This is shown in the simple schematic below. Evidently the resourcing people and tech increases with the maturity level too.



For Easy Property it is recommended that MLOps level 0 is selected. The figure and table to follow highlights the characteristic and challenges taken from Google’s “MLOps: Continuous delivery and automation pipelines in machine learning”¹¹ for this level.



MLOps level 0: Manual process for selection	
Characteristics	Challenges
Manual, script-driven, and interactive process: Each step, from data analysis to model training, is performed manually using experimental code in notebooks.	Lack of automation and CI: Manual execution slows down development, and there is no dedicated continuous integration (CI) process.
Disconnection between ML and operations: Data scientists hand off trained models to engineers for deployment, requiring additional steps for production readiness.	Infrequent release iterations and limited adaptability: Models are rarely updated or retrained, hindering their ability to adapt to changes in the real world.
Deployment focuses on prediction service: The process emphasizes deploying the trained model as a prediction service rather than the entire ML system.	Insufficient performance monitoring and tracking:

	Lack of active monitoring makes it difficult to detect model performance degradation and behavioural drifts.
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As this is a first initiative in ML for Easy Property the aim is to manage both risk and reward. Expanding the investment to level 1 and 2 would be advised only if Easy Property had:

1. A high variability in the data sets requiring frequent retraining.
2. A greater financial or public risk on getting its machine learning predictions wrong i.e., providing credit scores or assessing images for cancer.
3. The strategic objective to expand their service into more ML projects.

It is therefore advised that the existing challenges would be manageable for Easy Property however further steps can be taken based on the cost savings from this proposed project to move to the next level.

Conclusion

Easy Property will be able to save almost \$2.5 Million by delivering on the proposed CRISP-ML Project Plan and implementing a Level 0 – MLOps operating model. Not only will they double their money back, but they will reduce the manual effort of staff by a minimum of 75%. This will boost employee morale and reduce resourcing cost.

For their clients it will cut the turnaround time to upload photos by a minimum of 50% and allow for quicker scalability and downsizing as the selling market fluctuates through economic cycles. This may also provide greater competitive advantage and cost management.

However, with ineffective compliance and management of these project phases and mentioned risks there is an 85% chance of failure⁵. Therefore, it is critical that there is effective sponsor support, project management, stakeholder engagement, domain expertise, company buy-in and governance to successfully implement the proposed CRISP-ML Project Plan and Level 0 – MLOps operating model.

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